The Self-Improvement Ceiling: Why External Verification and Human-Anchored Data Remain the Scarce, Defensible Input in a Synthetic-Data Economy

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Abstract

A seductive narrative holds that large language models can bootstrap themselves out of the data wall: generate their own training data, judge their own outputs, and improve without bound. We assemble the empirical record against this view and show it imposes a hard ceiling. Imitating a stronger model transfers style but closes "little to none" of the real capability gap [1]. Self-rewarding improves AlpacaEval 2.0 win rate from 9.94% to 20.44% over three iterations but the authors concede the effect "likely saturates in realistic scenarios" because the model judges itself [2]. Expectation-maximization self-training (ReST-EM) lifts PaLM-2-L on MATH from 44.02% to 50.96% — but only by filtering samples with an external binary correctness signal, and most gains arrive in iteration 1 [3]. Pure recursion without a real-data anchor collapses irreversibly, erasing distribution tails (OPT-125M perplexity degrades from a 34 real-data baseline as generations compound) [4]. Even RLVR's headline gains are partly elicitation: base models match or overtake RLVR models at large k (Qwen-2.5-7B MATH500 pass@256 96.2% base vs 97.2% RLVR; on LiveCodeBench base ~50% vs RLVR 42.8%) [5]. The common thread: every durable gain requires an external anchor — verifiable rewards or human-grounded data. We argue this anchor is the structurally scarce input, validated by a market where one lab discussed >$1B/year on RL environments [6,7] and by 2025-2026 evidence that scaling the collection of verifiable environments — not RL compute — drives further gains (+3.37% from 400 environments vs +0.49% from 3x more compute [17]; RL performance follows a finite sigmoidal compute asymptote [16]), and derive implications for proprietary vertical data strategy.

1. Introduction

The single most consequential question in AI data strategy for 2026 is whether self-improvement makes training data free. If a model can generate its own data, grade its own outputs, and iterate, then the looming exhaustion of public human text — forecast for 2026-2032 [8] — becomes irrelevant, and any investment thesis premised on the scarcity of human-grounded data collapses. This paper argues, on the empirical record, that the opposite is true: every durable self-improvement gain documented in the literature depends on an external anchor — either a verifiable reward signal or real human-grounded data — and that anchor is precisely the input that does not become abundant in a world flooded with synthetic data.

The argument matters because the four phenomena that look like 'free data' — model imitation/distillation, self-rewarding, expectation-maximization self-training, and recursive synthetic generation — are routinely cited individually as evidence that bootstrapping works. Read individually, each sounds like incremental progress toward self-sufficient AI. Read together, they impose a hard ceiling. Imitation mimics style, not capability [1]. Self-rewarding is bounded by the model judging itself [2]. Self-training that works smuggles in an external correctness filter [3]. Pure recursion collapses [4]. The synthesis — that external verification or human grounding is mandatory, making that anchor the durable scarce good — is the inversion this paper develops and quantifies.

We contribute: (i) a unified, number-grounded account of the self-improvement ceiling across four mechanisms; (ii) new 2024-2026 evidence — including the RLVR pass@k crossover [5] — that even reinforcement learning on verifiable rewards is partly elicitation of latent skill rather than creation of new skill; (iii) a market validation showing that frontier labs are already pricing the external anchor as a scarce asset (>$1B/year discussed by a single lab on RL environments [6,7]), reinforced by 2025-2026 evidence that scaling the collection of verifiable environments — not RL compute alone — is what drives further gains [17]; and (iv) an explicit data-strategy framework for where created, human-anchored, verifiable data is most defensible.

2. Background and Related Work

The backdrop is the 'data wall.' Villalobos et al. [8] estimate that LLMs will be trained on datasets roughly equal in size to the available stock of public human text data between 2026 and 2032 (or slightly earlier if models are overtrained). Repetition does not rescue the situation: up to ~4 epochs of repeated data yields negligible change versus unique data, and with more repetition the value of added compute eventually decays to zero [9]. This creates structural demand for a non-text-stock source of fresh gradient signal.

Three candidate escape routes have been proposed. The first is distillation/imitation: train a weaker model on a stronger model's outputs. The second is self-rewarding: let the model act as its own reward model (LLM-as-a-Judge) and train on its own preference data [2]. The third is self-training with an external filter, formalized as expectation-maximization in ReST-EM [3], and its RL generalization, reinforcement learning with verifiable rewards (RLVR). Cutting across all of these is the model-collapse literature [4], which characterizes what happens when the synthetic loop runs without a real-data anchor.

Our work is closest in spirit to insight i04 (the model-collapse moat) but addresses the demand side: i04 argues real data resists collapse; we argue self-improvement loops actively require an external anchor to function at all, making that anchor scarce by construction. We also extend the 2025 RLVR-boundary debate [5,10] into a data-strategy conclusion that neither paper draws.

3. Methodology and Approach

This is an evidence-synthesis paper, not a new training run. Our method is adversarial number extraction: for each of the four mechanisms, we locate the strongest published result in favor of self-sufficiency, copy its headline number verbatim from the primary source, and then locate the same paper's own stated boundary condition or the counter-result that bounds it. A mechanism counts as 'ceilinged' only when the boundary is stated by the proponents themselves or by a direct replication, not by a critic.

Inclusion criteria: (i) the result must be quantitative and traceable to a primary arXiv/Nature source; (ii) numbers are copied exactly, with benchmark and model named; (iii) projections (e.g., market size) are labeled as estimates, never asserted as empirical. We deliberately weight self-stated limitations — e.g., the Self-Rewarding authors' own concession that the effect 'likely saturates in realistic scenarios' [2] — because a proponent conceding a ceiling is stronger evidence than a critic asserting one.

We define the central quantity: an external anchor is any signal whose correctness is determined outside the model's own parameters — a unit test, a numeric answer key, a coding/claims adjudication, a human label, or experimentally-resolved ground truth. The thesis is operationalized as a falsifiable claim: if any of the four mechanisms produced durable, compounding capability gains without an external anchor, the thesis is false. We then check each mechanism against this claim. Finally, we triangulate the economic prediction (the anchor is scarce and priced) against observed 2025-2026 market behavior [6,7,11], treating market data as corroborating, not load-bearing, evidence.

4. Evidence I: Imitation Transfers Style, Not Capability

The cleanest case against 'free data via distillation' is Gudibande et al. [1]. Finetuning open base models on outputs from a stronger proprietary model (ChatGPT) produced imitation models that crowd workers initially rated as competitive with ChatGPT. But targeted automatic evaluation found they 'close little to none of the gap from the base LM to ChatGPT on tasks that are not heavily supported in the imitation data' [1]. The imitation models are 'adept at mimicking ChatGPT's style but not its factuality.'

The mechanism is diagnostic: human raters reward fluency and format — exactly what imitation transfers — while the underlying factual and reasoning capability, which lives in the teacher's training data and scale, does not move. The paper's prescription is telling for data strategy: the highest-leverage action is improving the base model itself, i.e., investing in better data and pretraining, not in cheaper imitation. Distillation redistributes capability that already exists somewhere in a real-data-trained teacher; it does not manufacture new capability from nothing.

Implication: a synthetic-data economy built on distilling stronger models is a closed system that cannot exceed the real-data-trained frontier that seeded it. The capability ceiling of every distilled model traces back to a model that was, in turn, trained on human-grounded data.

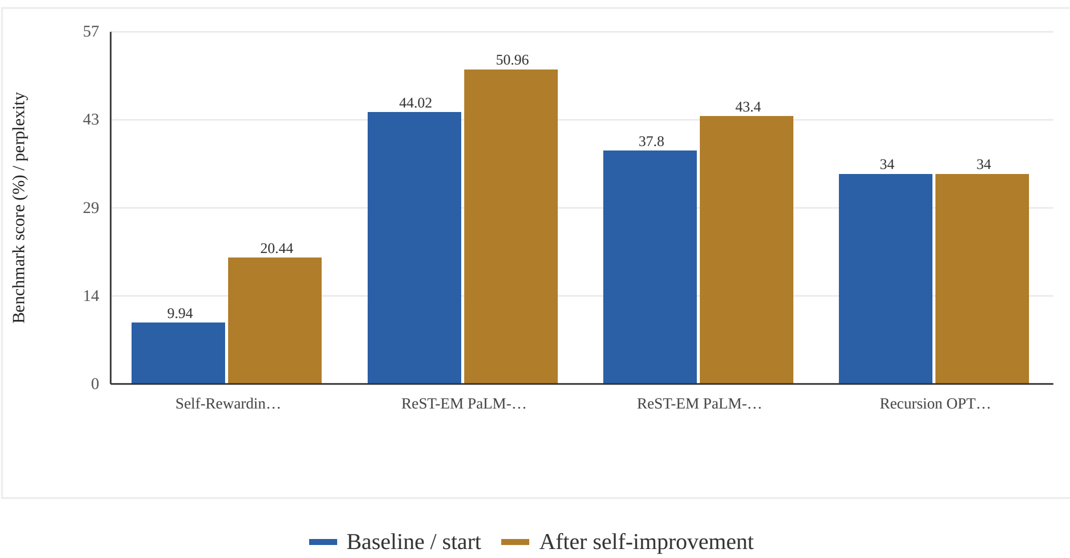

Figure 1. The Self-Improvement Ceiling: Headline Gain vs. Stated Boundary, by Mechanism. Each mechanism's strongest published gain shown against its endpoint. Self-rewarding rises 9.94%->20.44% but the authors state it 'likely saturates in realistic scenarios' (internal judge) [2]. ReST-EM rises 44.02%->50.96% on MATH and 37.80%->43.40% on APPS Introductory, but ONLY via an external correctness filter, with most gain in iteration 1 [3]. For recursion, the verified primary-source value is the 34-perplexity real-data baseline (from a 115 zero-shot baseline); without a real-data anchor, perplexity degrades across generations and distribution tails vanish (model collapse) — the recursion bar is held at the verified 34 baseline because the specific degraded endpoint is reported graphically, not as a single text value, in the primary source [4]. Mixed units; bars are within-mechanism, not cross-comparable; perplexity is lower-is-better. *[2] Yuan et al. 2024 (arXiv:2401.10020, Table 1); [3] Singh et al. 2023 (arXiv:2312.06585); [4] Shumailov et al. 2024 (Nature 631:755-759), OPT-125M/WikiText-2 baseline perplexity 34. · Empirical data.*

Table 1. Each row pairs the strongest published evidence for self-sufficiency with the same literature's stated boundary. In every case a durable gain requires an external anchor (verifiable reward or real data).

Mechanism	Strongest pro-bootstrap result (verbatim numbers)	Stated boundary / external anchor required	Source
Imitation / distillation	Imitation models initially rated competitive with ChatGPT by crowd workers	Close 'little to none' of capability gap on tasks not in imitation data; 'adept at mimicking ChatGPT's style but not its factuality'. Anchor = the real-data-trained teacher	[1]
Self-rewarding (LLM-as-Judge)	Llama-2-70B AlpacaEval 2.0 win rate 9.94%->15.38%->20.44% over 3 iters; beats Claude 2, Gemini Pro, GPT-4 0613	Authors: improvement 'likely saturates in realistic scenarios'; bounded by model judging itself (internal anchor)	[2]
EM self-training (ReST-EM)	PaLM-2-L MATH 44.02%->50.96%; PaLM-2-S* APPS Intro 37.80%->43.40%; surpasses human-data finetuning	Requires external scalar/binary correctness filter; 'most of the gains for APPS come from the first iteration,' then regresses	[3]
Recursive synthetic generation	(none — limiting case)	Irreversible model collapse; tails of the distribution vanish; OPT-125M real-data baseline perplexity 34 degrades across generations. Only minor degradation if 10% real data retained	[4]

[1] arXiv:2305.15717; [2] arXiv:2401.10020; [3] arXiv:2312.06585; [4] Nature 631:755-759 (2024) / arXiv:2305.17493.

Table 2. RLVR wins at low sampling budget but the base model catches up or overtakes at high k, indicating RLVR concentrates on already-reachable solutions rather than creating new capability. The capability ceiling is set by the base model's human-grounded training data; large-scale RL-compute studies independently find RL performance approaches a finite sigmoidal asymptote [16], and the frontier gains accrue from scaling verifiable environments rather than compute [17].

Model / Benchmark	Base pass@1	RLVR pass@1	Base pass@large-k	RLVR pass@large-k	Winner at large k
Qwen-2.5-7B / MATH500	34.5%	74.4%	96.2% (k=256)	97.2% (k=256)	~Tie (gap nearly closed)
Qwen-2.5-7B / LiveCodeBench	23.8%	28.1%	~50% (k=256)	42.8% (k=256)	Base model
Qwen-2.5-32B / Minerva	(lower)	(higher)	base +~9% at k=128	—	Base model

Yue et al. 2025, arXiv:2504.13837, MATH500 / LiveCodeBench / Minerva pass@k results (verified via arXiv HTML).

Table 3. Frontier labs and markets are pricing externally-grounded verifiable signal as a scarce, defensible asset, with an explicit exclusivity premium for non-replicable (proprietary) anchors.

Metric	Reported value	Note	Source
Single-lab annual RL-environment spend (discussed)	> $1,000,000,000	Anthropic, reported Sep 2025 (via The Information); 'discussed', not confirmed spend	[6,7]
RL-environment contract size	6-7 figures per quarter	Per-environment	[7]
Exclusivity premium	~4-5x non-exclusive price	Direct evidence proprietary anchors are the moat	[7]
Complex environment replica (e.g. Slack)	~$300,000	Simpler website replicas ~$20k each	[7]
Per-task pricing	~$200-$2,000 (up to ~$20k for hard SWE)	Task-level pricing	[7]
Legacy data-labeling market	~$5B, growing 50%+ YoY	~3 $1B+ players (Scale, Mercor, Surge)	[11]
RL-env startup field -> consolidation	Many today -> 3-5 winners projected	Labs concentrate spend among trusted partners	[11]
Environment scaling vs compute scaling (RL gains)	+3.37% (400 verifiable envs) vs +0.49% (3x compute)	Owning more verifiable environments — not more compute — drives RL gains; the anchor is the asset	[17]

[6] TechCrunch 2025-09-21 (citing The Information); [7] Epoch AI 'An FAQ on RL Environments'; [11] Wing VC RL-environment market analysis. Reported/estimated figures, not audited. [17] RLVE (arXiv:2511.07317): +3.37% from 400 verifiable environments vs +0.49% from 3x more RL compute.

5. Evidence II: Self-Rewarding Is Bounded by the Judge

Self-Rewarding Language Models [2] is the strongest published case for self-sufficiency: a Llama-2-70B model generates its own responses, scores them as its own LLM-as-a-Judge, builds preference data, and trains via Iterative DPO. The results are real and improve monotonically across three iterations on AlpacaEval 2.0 (win rate vs GPT-4 Turbo): M1 (SFT baseline) 9.94%, M2 15.38%, M3 20.44% [2]. After three iterations the model outperforms many existing systems on the AlpacaEval 2.0 leaderboard, including Claude 2, Gemini Pro, and GPT-4 0613.

But the boundary is stated by the authors themselves: the improvement 'likely saturates in realistic scenarios' [2]. The structural reason is that the reward model and the policy are the same network — the system can only reward what it can already recognize as good. As the policy improves, its own judgments become the binding constraint; it cannot reliably reward capability it does not yet possess. The paper tests only three iterations and explicitly flags the scaling behavior of more iterations as open. This is not a free lunch; it is a finite extraction of latent preference-discrimination ability the base model already held.

The contrast with externally-anchored methods is the crux. Self-rewarding rises and then plateaus because the anchor (the judge) is internal. The moment the signal source is external and verifiable, the ceiling moves — which is exactly what the next section shows.

6. Evidence III: Self-Training Works Only With an External Correctness Filter

ReST-EM [3] is the proof that self-generated data can genuinely help — and the proof that it requires an external anchor. The method is expectation-maximization: generate many samples, keep only the correct ones (filtered by a binary correctness signal), finetune on survivors, repeat. On the Hendrycks MATH benchmark, PaLM-2-L improves from a base 44.02% to 50.96% pass@1; on APPS (Introductory) coding, PaLM-2-S* improves from 37.80% to

43.40% [3]. The authors state plainly that the method 'significantly surpasses fine-tuning only on human data' on these verifiable tasks.

The load-bearing detail is the filter. ReST-EM works specifically on 'tasks where we have access to scalar feedback, for example, on math problems where one can verify correctness' — in practice a binary correct/incorrect verdict from an answer key or a unit test [3]. Remove that external signal and the method has no way to separate good self-generated samples from bad ones; it degenerates into the recursive loop that collapses (Section 7). The self-generation is free; the verification is not.

Two further facts sharpen the point. First, gains concentrate in iteration 1: 'most of the gains for APPS come from the first iteration, and performing more iterations leads to a regression in performance,' and MATH test improvements are 'small after the first iteration' [3]. The train-test gap widens with iterations — i.e., self-training extracts a finite reservoir, then begins to overfit. Second, this is RLVR in miniature: ReST-EM is EM self-training filtered by verifiable reward, the same recipe that scaled into models like DeepSeek-R1-Zero [13] and, by late 2025, into the open frontier — scaling RL post-training compute brings DeepSeek-V3.2 to parity with GPT-5, and its high-compute Speciale variant won gold at the 2025 IMO and IOI [18]. The verifiable signal — not the self-generation — is doing the work, and it is the verifiable environment, not the algorithm, that the recipe is hungry for.

7. Evidence IV: Without a Real-Data Anchor, Recursion Collapses

The fourth mechanism is the limiting case: what happens with no external anchor at all. Shumailov et al. [4], published in Nature (2024), trained successive generations of a model on the previous generation's outputs. The result is 'model collapse': indiscriminate use of model-generated content in training causes irreversible defects in which the tails of the original content distribution disappear, and rare and unique elements vanish. The effect is universal across VAEs, Gaussian mixtures, and LLMs.

The numbers are concrete. Finetuning OPT-125M on real WikiText-2 reached a mean perplexity of 34 (from a zero-shot baseline of 115); when successive generations are trained only on the previous generation's model-generated data, perplexity degrades steadily across generations as the distribution's tails are truncated [4]. The mechanism is that low-perplexity (high-probability) samples accumulate across generations while the tails — the rare, informative regions — are progressively cut off. Critically, the documented mitigation is an external anchor: preserving 10% of the original real data during training leads to only minor degradation of performance across ten generations, rather than collapse [4].

This is the demand-side mirror of imitation (Section 4). Imitation cannot exceed the real-data frontier; recursion cannot even maintain it without a real-data anchor. Together they bracket the synthetic-data economy: it can redistribute and locally polish capability, but it cannot create or even sustain capability without a continuous injection of externally-grounded signal.

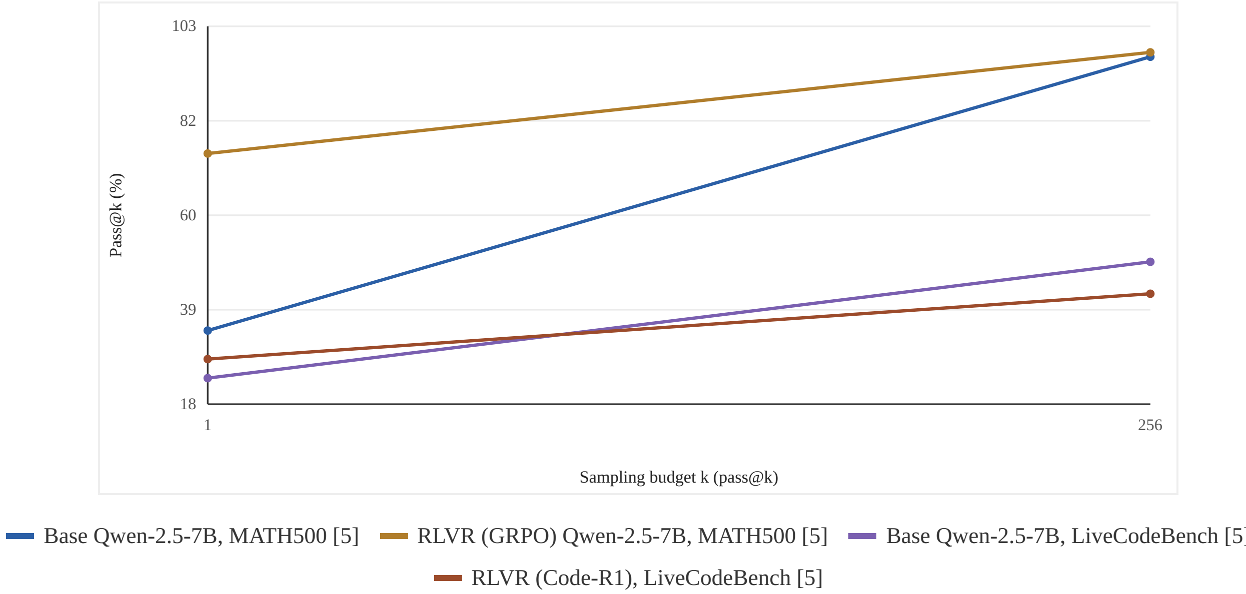

Figure 2. RLVR Elicits, It Does Not Create: Base Model Catches Up at Large k. RLVR wins decisively at k=1 (74.4 vs 34.5 on MATH500) but the base model nearly closes the gap by k=256 (96.2 vs 97.2), and on LiveCodeBench the base model OVERTAKES the RLVR model at large k (~50.0 vs 42.8). This is the signature of elicitation: RLVR concentrates probability on solutions the base model could already reach, rather than expanding the reachable set. The ceiling traces back to the base model's (human-grounded) training data — and large-scale RL-compute studies independently find RL performance approaching a finite sigmoidal asymptote [16]. *[5] Yue et al. 2025, 'Does RL Really Incentivize Reasoning Capacity Beyond the Base Model?' (arXiv:2504.13837), MATH500 and LiveCodeBench pass@k results (verified via arXiv HTML). · Empirical data.*

8. Analysis: Even RLVR Is Partly Elicitation, Not Creation

A natural objection is that RLVR — the verifiable-reward recipe behind frontier reasoning models — breaks the ceiling. The 2025 evidence complicates this. Yue et al. [5] show that RLVR-trained models beat their base models at small k but base models catch up or overtake at large k. On Qwen-2.5-7B / MATH500: base pass@1 34.5% vs RLVR 74.4%, but base pass@256 96.2% vs RLVR 97.2% — the gap nearly closes. On LiveCodeBench: base pass@1 23.8% vs RLVR 28.1%, but base pass@256 ≈50% vs RLVR 42.8% — here the base model overtakes [5]. On Qwen-2.5-32B / Minerva the base 'outperforms the RL-trained model by approximately 9% at k=128' [5].

The interpretation: RLVR often sharpens the sampling distribution toward solutions the base model could already reach, rather than expanding the reachable set. This is consistent with the spurious-rewards finding that RLVR can improve Qwen-2.5-Math-7B on MATH-500 by 21.4 points even with randomly assigned rewards — but that these gains fail to transfer to Llama-3 or OLMo-2, because the latent skill is model-dependent [10]. Counter-evidence exists: prolonged RL can uncover genuinely novel strategies [10], so the debate is live. The first large-scale (>400,000 GPU-hours) study of RL compute scaling sharpens the picture: RL performance follows a sigmoidal compute curve toward a finite asymptote, and most design choices (normalization, curriculum, off-policy algorithm) only modulate compute efficiency without moving that asymptote — only the recipe itself shifts the ceiling [16]. In other words, more RL compute reaches a bound; it does not abolish one. But the data-strategy conclusion is robust to either side of the debate: whether RL teaches or merely elicits, the ceiling traces back to what the base model absorbed from its (human-grounded) training data, and no synthetic loop can lift that ceiling [4,16].

The strategic inversion follows. If RLVR mostly elicits, then verifiable-reward data adds the most genuine new capability precisely where the skill is not already latent — in novel, specialized, under-documented domains absent from pretraining. The highest-value external anchor is therefore not math/code (where gains may be elicitation and the corpus is public) but proprietary vertical workflows where the capability genuinely does not yet exist in any base model. The 2025 environment-scaling result makes the point quantitatively: from a strong 1.5B model, jointly training across 400 verifiable environments yields +3.37% average across six reasoning benchmarks, whereas continuing the original RL run with 3x more compute yields only +0.49% — it is the breadth of verifiable environments, not the compute, that buys new capability [17].

9. Implications for Data Strategy

The four mechanisms converge on one prescription: own the external anchor. Distillation, self-rewarding, EM self-training, and recursion all either require an externally-grounded signal or collapse without one. The scarce, defensible input is therefore not raw tokens (which synthetic generation makes cheap) but verifiable, human-grounded signal — answer keys, adjudicable outcomes, real ground truth — that the loop cannot manufacture for itself.

This reframes 'environments with checkable outcomes' from a training trick into a manufacturable, durable data asset — and the 2025 frontier now optimizes for exactly this, with environment scaling (not compute scaling) shown to be the binding driver of further RL gains [17]. Three properties make it defensible. (i) Non-substitutability: synthetic data structurally cannot replace it (Sections 4, 7). (ii) Recurrence: like contamination-prone evaluations, verifiable environments must be refreshed as models saturate them, making them a subscription surface rather than a one-time dump. (iii) Vertical concentration: the richest verifiable, proprietary outcomes sit inside specialized workflows (clinical coding/claims that pass or fail, insurance adjudication, lab/diagnostic results), which are exactly the web-underrepresented tails where the marginal externally-anchored token is worth most and where base models lack latent skill (Section 8).

For a builder, the operational filter is two-axis: prioritize verticals scoring high on (1) a proprietary or hard-to-assemble data source and (2) an objective, checkable outcome. Math and code score high on axis 2 but low on axis 1 (public corpus, likely elicitation). Clinical claims/coding, regulated-industry workflows, and lab/test data score high on both — and are where created, human-anchored, verifiable data compounds into a moat rather than a commodity.

10. Investment Thesis and Market Validation

The thesis predicts that as self-training scales, frontier labs will pay escalating premiums for external anchors they cannot self-generate. The 2025-2026 market is consistent with this. In September 2025, Anthropic was reported (via The Information, relayed by TechCrunch) to have discussed spending over $1 billion on RL environments over the following year [6,7]. Per-environment economics are now observable: individual environment contracts run from six to seven figures per quarter, task pricing spans roughly $200-$2,000 (occasionally up to $20k for hard software-engineering tasks), and — critically for the moat thesis — exclusive deals command roughly a '4-5x' premium over non-exclusive ones [7]. Replicating a complex product environment (e.g., Slack) can cost ~$300k, versus ~$20k for simpler website replicas [7].

The market structure mirrors the thesis. The legacy human-data-labeling market is roughly $5B today, growing in excess of 50% YoY, with roughly three $1B+ revenue players (Scale, Mercor, Surge) and more than five $100M+ players (Turing, Invisible, and others) [11]. The RL-environments layer is forming on top: a field of seed-to-Series-A startups today (e.g., Mechanize, Prime Intellect) that the analysis projects will consolidate to '3-5 significant winners' as labs concentrate spend among trusted partners [11]. The exclusivity premium [7] is direct market evidence that a proprietary, non-replicable anchor is the defensible asset.

The VC-relevant inference: the durable value accrues not to whoever runs the RL algorithm (open and commoditizing — state-of-the-art GRPO-successor systems such as DAPO are fully open-sourced [20], and fully autonomous pipelines now mint thousands of verifiable environments with no human annotation [19]) but to whoever owns the scarce, non-replicable external anchor in a defensible vertical. Public-data anchors (math, code) will commoditize first; proprietary, regulated-vertical anchors with checkable outcomes are where exclusivity premiums — and therefore margins — persist.

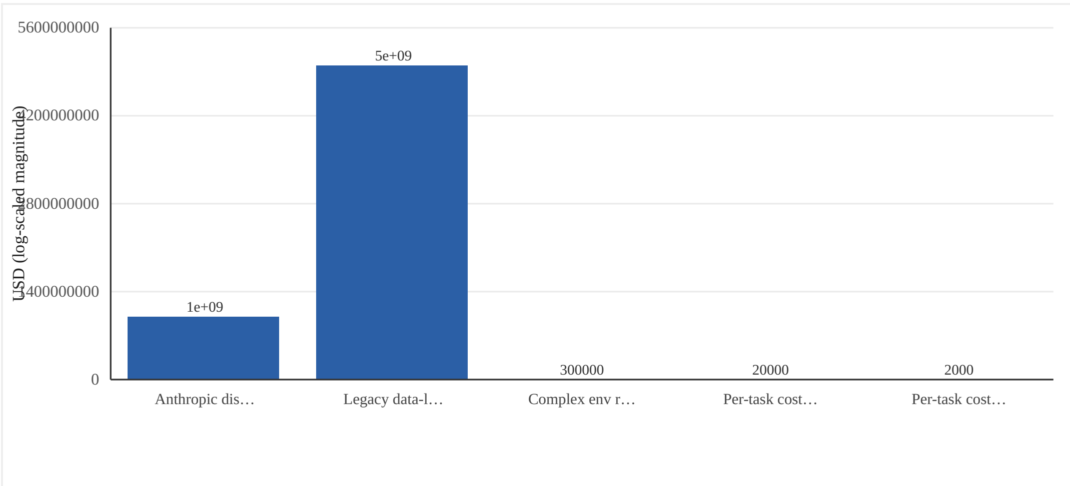

Figure 3. The Market Prices the External Anchor: RL-Environment Economics, 2025. Frontier labs are already pricing externally-grounded, verifiable signal as a scarce asset. A single lab discussed >$1B/yr on RL environments [6,7]; the legacy human-labeling market is ~$5B growing 50%+ YoY [11]; and exclusive environment deals carry a ~4-5x premium over non-exclusive [7], direct evidence that a non-replicable anchor is the defensible asset. Values are reported/estimated, not audited. *[6] TechCrunch 2025-09-21 (citing The Information); [7] Epoch AI, 'An FAQ on RL Environments' (epoch.ai); [11] Wing VC, 'Who Will Win the RL Environment Market'. Log-magnitude for visualization. · Empirical data.*

11. Limitations and Counter-Evidence

Several caveats bound this thesis. First, the saturation of self-rewarding [2] is the authors' projection, not a demonstrated asymptote — only three iterations were tested, and meta-rewarding variants and stronger base models may push the ceiling higher before plateauing. Second, the RLVR pass@k crossover [5] is contested: prolonged RL has been argued to uncover strategies inaccessible to the base model even under heavy sampling, so 'RLVR is pure elicitation' is too strong; our argument relies only on the weaker, robust claim that the ceiling traces to base-model data — a claim now reinforced by large-scale evidence that RL compute approaches a finite sigmoidal asymptote rather than escaping it [16]. The same scaling study shows the recipe can shift the asymptote, so the ceiling is not fixed for all time; our claim is the directional one that synthetic self-loops without an external anchor cannot move it.

Third, model collapse [4] assumes indiscriminate full replacement of real with synthetic data; the same literature shows that retaining a fraction of real data (e.g., 10%) sharply limits degradation [4], so a managed synthetic-plus-anchor regime is viable — which strengthens rather than weakens the claim that the real anchor is the load-bearing component, but means 'synthetic data is worthless' would be a misreading. Fourth, the benchmark numbers cited span different models and tasks (PaLM-2, Llama-2, Qwen-2.5) and are not directly comparable across mechanisms; they are illustrative of within-paper effects, not a unified scale.

Fifth, the market figures [6,7,11] are reported/estimated, sourced to trade reporting and a VC analysis, not audited financials; the >$1B Anthropic figure was 'discussed,' not confirmed spend, and should be read as directional. Finally, the data-strategy conclusion identifies where externally-anchored data is most valuable (proprietary verticals with checkable outcomes) but does not establish absolute willingness-to-pay, regulatory/data-rights feasibility (e.g., HIPAA, resale rights), or supply availability for any specific vertical — those remain to be validated buyer by buyer.

12. Conclusion

The promise that AI will generate its own training data and improve without bound is, on the evidence, a promise with a ceiling. Imitation transfers style but not capability [1]; self-rewarding saturates because the judge is internal [2]; expectation-maximization self-training works only because it is filtered by an external correctness signal, and even then most gains arrive in the first iteration [3]; and pure recursion collapses irreversibly without a real-data anchor [4]. Even RLVR, the recipe behind frontier reasoning, is at least partly elicitation of latent skill, with base models matching or beating RLVR models at large sampling budgets [5,10].

The unifying conclusion is an inversion of the 'free data' narrative: every durable self-improvement gain requires an external anchor — a verifiable reward or human-grounded data — and that anchor is the structurally scarce input that retains value even as synthetic data floods the world. The rise of self-training does not erode the value of the anchor; it reinforces it, because self-training cannot function without it.

For data strategy, this turns 'verifiable environments in proprietary verticals' from a niche tactic into the defensible asset class of the post-data-wall era. The market is already pricing it: exclusive RL-environment deals command ~4-5x premiums [7] and a single lab discussed over $1B in annual spend [6,7]. The winning move is to own the scarce external anchor — proprietary data crossed with checkable outcomes — in verticals where base models have no latent skill to elicit.

INVESTMENT THESIS

Self-improvement has a hard ceiling, and that ceiling is the whole investment case. Every 'free data' loop either needs an external anchor (ReST-EM's binary correctness filter; self-rewarding's quickly-saturating internal judge, 9.94%->20.44% then 'likely saturates') or collapses without one (recursion: OPT-125M perplexity degrades from a verified 34 real-data baseline as generations compound, with tails vanishing). Even RLVR is mostly eliciting latent skill — base models catch up or overtake at large k (LiveCodeBench: base ~50% vs RLVR 42.8% at pass@256; MATH500: 96.2% vs 97.2%) — which means new capability requires data the base model never saw. The defensible asset is therefore the scarce external anchor: proprietary data crossed with checkable outcomes, in verticals where base models have no latent skill to elicit (clinical claims/coding, regulated workflows, lab/diagnostic data) rather than public math/code. The market already prices this: a single lab discussed >$1B/yr on RL environments, contracts run 6-7 figures/quarter, and exclusive (non-replicable, proprietary) deals command a ~4-5x premium — the clearest signal that owning the anchor, not the algorithm, is where margin and moat live. The 2025-2026 frontier confirms the mechanism: scaling the *collection of verifiable environments* (not RL compute) is the binding driver of new capability (+3.37% from 400 environments vs +0.49% from 3x more compute), and RL compute itself approaches a finite sigmoidal asymptote — so the moat is the environment/anchor, not the algorithm (open systems like DAPO are fully released, and autonomous pipelines now mass-produce environments). Underwrite teams that lock exclusive rights to verifiable vertical data; discount anyone selling synthetic-data 'self-improvement' with no external anchor.

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